



END-SEMESTER PRESENTATION · MAY 2026

Zero-Shot Video Anomaly Detection using Vision-Language Models with Domain-Adaptive Verbalized Prompting

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Problem Statement & Our Approach

Why does zero-shot anomaly **detection** matter — and how does CLIP address it?

⚠ The Challenge

Traditional AD requires training data **per category**

Outputs only scores — **never explains** what is wrong

No temporal awareness for video streams

No method is zero-shot + explainable + temporal **simultaneously**

✦✦ Our Approach: CLIP Zero-Shot

Use CLIP's **joint vision-language embedding** space

Score = **cosine similarity** (damaged vs good text prompts)

No training data needed — zero-shot by design

Extensible to video via **temporal adapters**

At a glance: the left side highlights the limitations of conventional anomaly detection, while the right side shows how CLIP provides a **zero-shot, explainable, and scalable** path forward.

Image Input

CLIP ViT-L/14

Prompt Ensemble

Anomaly Score

Each stage contributes to a lightweight inference pipeline that can extend from static images to video.

Literature: PatchCore & WinCLIP

Foundational image anomaly detection methods — our primary baselines



PatchCore · CVPR 2022

- Patch-level features in a **memory bank**
- Anomaly = max distance from normal patches
- **99.1% AUROC** on MVTec AD
- Requires category-specific training; no explanations



WinCLIP · CVPR 2023

- First CLIP-based **zero-shot** AD method
- **Compositional Prompt Ensembles** + sliding window localisation
- **91.8% AUROC** zero-shot on MVTec AD
- Hand-crafted prompts; no defect explanations

 Our baseline: **88.5% zero-shot AUROC** using prompt ensembles, no sliding window — kept simple for video extensibility.

AnomalyCLIP, AnomalyGPT & MMAD

Advanced VLM approaches — and why a research gap persists



AnomalyCLIP · ICLR 2024

- Learnable **object-agnostic prompts**
- **~94% AUROC** on MVTec AD
- Better cross-domain generalisation
- Requires fine-tuning — **not zero-shot**
- No video or temporal capability



AnomalyGPT · AAAI 2024

- First MLLM for industrial AD
- **94.1% AUROC** with 1-shot
- Natural language **explanations** of defects
- Requires fine-tuning — not zero-shot
- No temporal support



MMAD · ICLR 2025

- Benchmark: **7 AD subtasks**
- Tests GPT-4o, Claude, LLaVA
- Even GPT-4o: **moderate accuracy only**
- Prompt design is critical for AD
- MLLMs alone are **insufficient**



Research gap validated: No existing method is simultaneously zero-shot, explainable, and temporally aware.

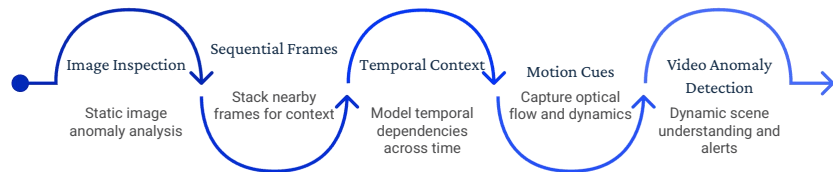
Pivot to Video AD

Why we extended from image anomaly detection to video anomaly detection



Why extend to video?

- Image AD: static, controlled environments
- Video AD: dynamic scenes with **temporal context**
- Professor's guidance: *"Look into domain adaptability beyond static images"*



[OPEN-VOCABULARY](#)[VIDEO AD](#)[SURVEY](#)

OVVAD & VAD Survey: Shaping Our Direction

CVPR 2024 + ACM 2025: zero-shot framework + survey evidence.

OVVAD Architecture

- **Frozen CLIP:** fixed ViT-B/16 features
- **Temporal adapter:** learns frame sequence cues
- **Text injection:** normal/anomaly descriptions
- **Anomaly synthesis:** boosts robustness

86.40% AUC on UCF-Crime

Why the Survey Matters

- **400+ methods:** broad VAD map
- **Key gaps:** weak generalization, weak time modeling, label-heavy
- **Open-vocab VAD:** handles unseen anomalies via language
- **Why DA-ZVAD:** supports zero-shot, language-aware direction

CVPR 2024

CVPR 2025

LAVAD & VERA — Training-Free Video AD

Two CVPR papers that complete our architectural puzzle

LAVAD

Caption-then-reason pipeline

- Frames → BLIP-2 Captions → LLM Scoring
- LLM reasons about **temporal events** across frames
- Dynamically adjusts normality baseline
- Outperforms all unsupervised VAD methods



Limitations: Noisy/hallucinated captions; high LLM inference cost

VERA

Verbalized domain prompting

- *"A dimly lit warehouse with conveyor belts"*
- Cross-modal refinement reduces caption noise
- Zero training required; domain-adaptive via text alone
- Explainable outputs; cross-domain robust



Running person: Normal in an arena, **anomalous in a warehouse** — domain context matters.

Comparative Analysis

No existing method satisfies all five desiderata — **DA-ZVAD** targets this gap.

❏ **What matters most:** zero-shot, training-free, explainable, temporal, and domain-adaptive performance.

Method	Venue	Zero-Shot	Training-Free	Explainable	Temporal	Domain-Adaptive
PatchCore	CVPR '22	X	X	X	X	X
WinCLIP	CVPR '23	✓	✓	X	X	X
AnomalyCLIP	ICLR '24	✓	X	X	X	X
AnomalyGPT	AAAI '24	X	X	✓	X	X
OVVAD	CVPR '24	✓	X	X	✓	Partial
LAVAD	CVPR '24	✓	✓	Partial	✓	X
VERA	CVPR '25	✓	✓	✓	✓	✓
DA-ZVAD (Ours)	Proposed	✓	✓	✓	✓	✓

✓ **Key insight:** DA-ZVAD is the only proposed method satisfying all five desiderata simultaneously — zero-shot, training-free, explainable, temporal, and domain-adaptive.

CLIP Baseline — Results on MVTec AD

15 categories · 1,725 test images · Zero training data

88.5%

Mean AUROC

Zero-shot, no training

90.7%

F1 Score

Strong detection precision

135.8s

Inference Time

On T4 GPU, full dataset

⚠️ **Gap vs WinCLIP (91.8%):** attributed to omission of multi-scale sliding window — a deliberate trade-off for video extensibility.

Category Breakdown

Textures — 99.5% mean

Carpet 99.2% · Grid 99.7% · Leather **100%** · Tile 98.7% · Wood **100%**

Global appearance captured well by CLIP

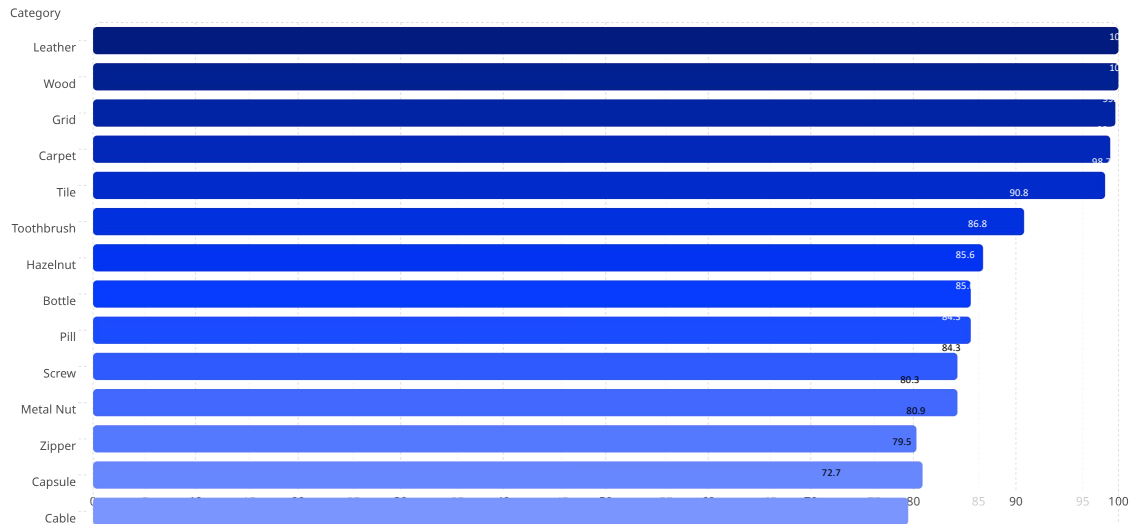
Objects — 83.1% mean

Bottle 85.6% · Cable 79.5% · Capsule 80.9% · Transistor 72.7% · Zipper 80.3%

Localised defects need stronger spatial reasoning

AUROC by Category

Visual performance breakdown across all 15 MVtec AD categories — textures consistently outperform objects.



Key pattern: texture categories are nearly perfect, while object categories are more sensitive to localized and structural defects.

Textures lead

Leather, wood, grid, carpet, and tile cluster at the top.

Objects lag

Transistor, cable, capsule, and zipper show the toughest cases.

Next phase

Improve spatial reasoning for fine-grained defect localization.

Texture categories (leather, wood, grid) achieve near-perfect scores. Object categories with localised or structural defects — transistor, cable, capsule — highlight the need for spatial reasoning enhancements in the next phase.

Video Extension — Per-Frame CLIP Scoring

Simulating conveyor belt quality inspection with pseudo-video frames

0.307

Normal Score

0.894

Anomalous Score

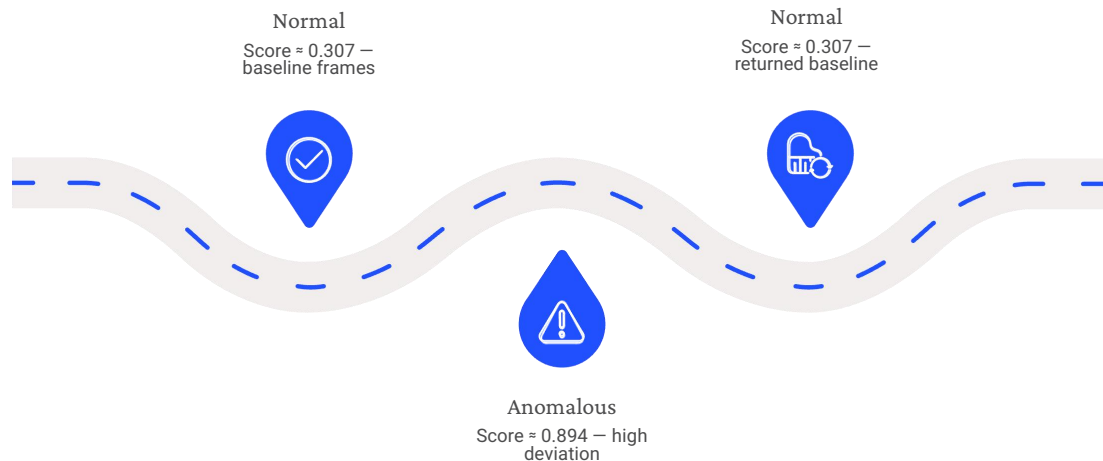
0.587

Score Gap

Experimental Setup

- MVTec bottle images sequenced as **pseudo-video frames**
- 30 frames: Normal $\times 10 \rightarrow$ Anomalous $\times 10 \rightarrow$ Normal $\times 10$
- Each frame scored **independently** by CLIP
- Simulates real-time conveyor belt inspection

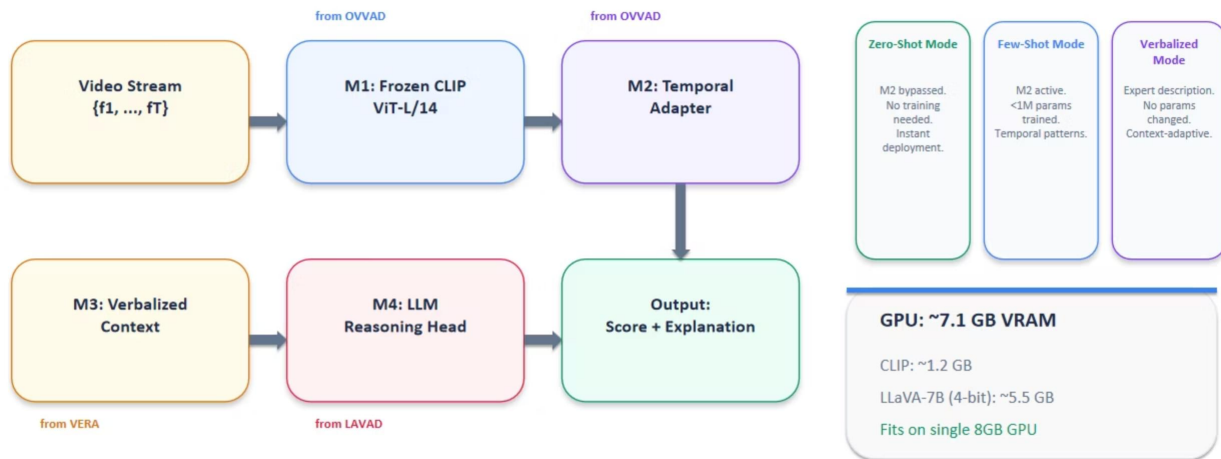
Frame Sequence — Score Pattern



Clean separation validates CLIP for video: A 0.587 score gap between normal and anomalous frames confirms CLIP's discriminative power as a foundation for the DA-ZVAD video pipeline.

DA-ZVAD — Proposed Architecture

A clean, end-to-end system that turns video input into an anomaly score and a natural-language explanation.



- ✓ **Feasibility:** Total VRAM ~7.1 GB (CLIP ViT-L/14 ~1.2 GB + LLaVA-7B 4-bit ~5.5 GB + Temporal Adapter ~0.2 GB + Frame Buffer ~0.2 GB). Runs on a single RTX 3060 (8 GB) or T4. No multi-GPU setup needed.

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Thank You

Questions & Discussion

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